

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: **Terra**

Project / Product Name: **Smart Guided Troubleshooting Engine / SureStep**

Organization / Institution (if any): **SRM Institute Of Science and Technology**

Submission Date: **30 September 2026**

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? **Yes**

AI was used in two ways. (1) As a development tool: Claude (Anthropic), through Claude Code and the Claude app, for design discussion, code, tests, documentation and review. (2) Inside the product: Google Gemini 3.1 Flash-Lite parses each new complaint in one call (temperature 0, fixed JSON schema), and the all-MiniLM-L6-v2 sentence-embedding model powers the semantic cache. Every AI-written change was checked by the automated test suite (263 tests) and by separate review passes, and the team ran and verified the results on its own machine before committing.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: **Claude was used to analyse Samsung's Theme 02 brief, compare approaches, and propose improvements. The Article Fit feature was proposed by Claude and chosen by the team.**

Code generation or assistance: **Most backend, frontend, test and evaluation code was written by Claude (Claude Code) to the team's specifications and a written rule file (CLAUDE.md). The team reviewed, ran and committed it.**

UI / UX design: **The React demo UI (customer view, guided mode, engineer view) was generated with Claude and checked screen by screen in a headless browser.**

Content creation: **README, limitations document, presentation text and the demo-video script were drafted with Claude. Performance figures are generated by the evaluation code, never written by AI or by hand.**

Data analysis: **Claude wrote the evaluation harness and the catalog-audit scripts; every figure is computed by that code from Samsung's supplied data.**

Testing / debugging: **Claude wrote regression tests and ran independent review passes, which found defects (for example a URL leak and a crash path) that were then fixed and locked by tests.**

Other: **AI inside the product: Gemini 3.1 Flash-Lite as a structured parser only (it cannot output a deeplink), and all-MiniLM-L6-v2 embeddings for the cache.**

4. Feature Origin Classification (Please add/delete based on the number of the feature)

Feature 1

1. Feature Name: **Complaint-to-plan extraction (one LLM call)**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude Code) for design and code; Google Gemini 3.1 Flash-Lite at runtime.

Prompt used: Representative: "Build the extraction stage: one Gemini call that returns steps, categories and source spans as JSON against a fixed schema, with no deeplink field."

Output summary: Extraction prompt, JSON schema, Gemini provider with retry, and an offline parser for testing.

Modification: Team set the rule that the model is a parser, not an authority. Model-reported spans were found wrong (0 of 29), so every span is relocated by code instead.

Feature 2

1. Feature Name: **Deterministic deeplink resolver (gates 0 to 6)**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude Code).

Prompt used: Representative: "Map each step to exactly one of the 578 catalog entries, or to the dummy link, or to null, with polarity, scope and label checks. Never let the model choose the URI."

Output summary: BM25 retrieval plus seven ordered gates, with an ablation that measures what each gate adds.

Modification: Gate order frozen by the team; a label-support check was added after the catalog audit, accepting a lower headline precision (99.2% to 98.7%) for more correct answers.

Feature 3

1. Feature Name: **Catalog audit and health report**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude Code).

Prompt used: Representative: "Find catalog entries whose own description contradicts their label, and generate a report Samsung's catalog team could act on."

Output summary: Audit module and generated report: 50 unsupported labels, 5 indistinguishable groups.

Modification: Findings verified against the supplied catalog; the resolver now refuses those entries.

Feature 4

1. Feature Name: **Semantic cache (exact and semantic tiers)**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude Code); all-MiniLM-L6-v2 embeddings at runtime.

Prompt used: Representative: "Serve reworded complaints about the same article from cache with no LLM call, and never return a plan built for a different article."

Output summary: Two-tier cache partitioned by article, plus a 10,000-scenario scale test.

Modification: The scale test exposed a 24-second fast path and a collapsing hit rate; both fixed and tested.

Feature 5

1. Feature Name: **Guided mode, safety gate and agent handoff**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude Code).

Prompt used: Representative: "Walk the customer through the validated plan one step at a time, confirm before every last-resort step, and hand an agent exactly what was tried."

Output summary: Guided session API, server-enforced confirmation gate quoting only the article's own warning, and a handoff built without any model call.

Modification: Review found warnings attached to the wrong steps; restricted to destructive steps and whole sentences from the article.

Feature 6

1. Feature Name: **Article Fit (does the article match the complaint?)**
2. Self-Generated/AI-Generated/Both: **AI-Generated idea and code; team-selected and verified**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude app, cloud session); Gemini 3.1 Flash-Lite at runtime.

Prompt used: "Isn't there something like a true feature, something I can add to my prototype that is different from what others built?" followed by "choose whichever is best according to you and can be called a real feature, not a side feature."

Output summary: Each problem the customer described is marked Covered, Not covered or Unverified; Covered counts only with a whole sentence quoted from the article. Shown to the customer, in guided mode and to the agent.

Modification: Two independent review passes found a URL leak, weak quote checking, a crash path and a cache issue; all were fixed with regression tests before submission.

Feature 7

1. Feature Name: **Evaluation harness and generated metrics**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude Code).

Prompt used: Representative: "Every number must be generated by code, never typed; measure compliance, precision, cost and cache behaviour on the supplied rows."

Output summary: Evaluation scripts that regenerate report.json and metrics.md.

Modification: Team rule: no figure in the deck or video is quoted unless the harness produced it.

Feature 8

1. Feature Name: **Demo UI, presentation and video script**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification)

AI tools / platform: Claude (Claude Code and Claude app).

Prompt used: Representative: "Build a demo UI with a customer view and an engineer view"; "write a professional pitch for the demo video."

Output summary: React + Vite UI, slide text on Samsung's template, and a timed run-of-show for the video.

Modification: Team fills team details, records and narrates the video, and checks every figure against metrics.md.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Ayush Garg

Role: Primary Member

Signature: Ayush Garg

Date: **30 September 2026**